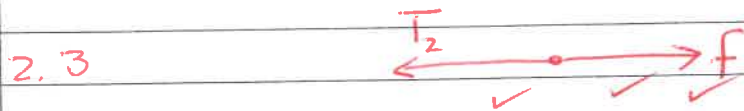


START NEW QUESTION ON NEW PAGE

$$2.1 \quad a = \frac{v_f - v_i}{\Delta t} = \frac{1,2 - 0}{0,5} = 2,4 \text{ m s}^{-2}$$

2.2 Newton 2nd law

$$2.4 \quad (B) \quad T_2 - f_f = ma$$

$$T_2 - 19,6 = 5(2,4) = 31,6 \text{ N}$$

$$2.6 \quad (A) \quad F_{\text{App}} - T_2 - f_f = ma$$

$$F_{\text{App}} - 31,6 - 3,9 = 1(2,4)$$

$$F_{\text{App}} = 37,9 \text{ N}$$

- 2.7.1 Force opposes movement when in contact
2. Not Fndt No friction
3. moves backwards inertia

$$2.7.4 \quad W = mg = N = 0,1 \times 9,8 = 0,98 \text{ N}$$

$$f_f = \mu F_N$$

$$= 0,4 \times 0,98 = 0,39 \text{ N}$$

$$S. \quad F_{\text{ndt}} = ma$$

$$0,39 = 0,1 a$$

$$a = 3,9 \text{ N}$$

Name : _____

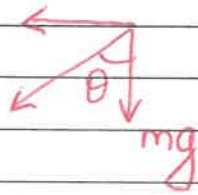
Teacher _____

START NEW QUESTION ON NEW PAGE

3.1 ~~E₁~~ ~~E₂~~ Force experienced per unit
+ charge. (p 12 EG) ✓ ✓ 2

3.2 $F = Eq$ ✓
 $420 \times 6 \times 10^{-6}$ ✓
 $= 2,52 \times 10^{-3} \text{ N}$ ✓ (3)

3.3



$\tan \theta = \frac{F_E}{mg} \checkmark = \frac{2,52 \times 10^{-3}}{0,003 \times 9,8 (0,0294)} \checkmark$ (4)
 $= 85,1^\circ$ ✓
 $4,89^\circ$ ✓

3.4 Bigger angle in opposite direction to ✓ (2)

35.1 $F = \frac{GM_1M_2}{r^2} \checkmark = \frac{6,67 \times 10^{-11} \times 6,4 \times 10^{23} \times 1,48 \times 10^5}{(3390 \times 1000 + 20,1 \times 10^3)^2} \checkmark$
 $= 1,15 \times 10^{14} \text{ N} \checkmark$ (5)

35.2 ~~A~~ C : ✓ Newton 3 ✓ (2)

START NEW QUESTION ON NEW PAGE

4. EMF is max work done/charge ✓✓

(2)

4.2 $V = IR$
 $10 \times 0,05$
 $= 0,5V$

(3)

4.3 $P = I^2 R$ $\therefore I = \sqrt{\frac{P}{R}}$
 $= \sqrt{\frac{55}{12}} = 2,14A$

(3)

4.4 $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
 $\frac{1}{12} + \frac{1}{12}$ $\therefore R = 6\Omega$

(2)

4.5 $EMF = I(R+r)$ ✓

$EMF = 10(0,05) + 10r$ ✓

$EMF = 4,28(6) + 4,28r$ ✓

$10(0,05) + 10r = 4,28(6) + 4,28r$

$10r - 4,28r = -0,5 + 25,68$

$5,72r = 25,18$

$r = 4,4\Omega$ ✓

(7)

$EMF = 10(0,05) + 10(4,4)$ ✓
 $= 44,5V$ ✓

4.6 Decrease
 $R \downarrow$

$EMF = I(R+r)$ $I \uparrow$

$EMF = IR + Ir$
 $\downarrow$
 $\downarrow$ $\downarrow$
 I r
 $\downarrow$ $\downarrow$
 $I \uparrow$ $V \uparrow$

(3)

$\therefore Pd \downarrow$

Energy $\propto Pd$ ✓

$Emf = I(R+r)$
 $\downarrow$

$P = I^2 R$